

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2025

Bachelor in Computer Applications

Course Title: **Probability & Statistics**

Code No: **CACS203** Semester: **III**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. Discuss the application of statistics in computer application.
3. The 100 salesmen employed by a company have booked the following number of orders for a newly introduced FAX machine during the last six months:
Number of orders Booked: 10 – 2020 – 3030 – 4040 – 5050 – 6060 – 7070 – 80
No. of salesman: 41225301586
Calculate mean, median and mode of the above data.
4. Define correlation. A school teacher believes that there is a linear relationship between the verbal test score (y) for eighth graders and the number of library books checked out (x). Following are the data collected on 8 students:
X: 121537105229Y: 7785485975419465
(i) Compute the correlation coefficients r between X and Y and Interpret it.
(ii) Find coefficient of determination and interpret it.
5. Define regression. The technician now varies the temperature ($^{\circ}\text{C}$) while keeping other conditions as constant as possible and obtain the following results:
Yield (Y): 127128130131133
Temperature (X): 7075808590
(i) Construct the regression line of yield on temperature
(ii) Predict the yield when temperature is 95.
(iii) Interpret the regression coefficients.
6. Define probability. The odds against of A solving a problem as 8 to 6 and the odds in favor of B solving the same problem are 14 to 10. What is the probability that (i) Both A and B will solve it and (ii) A solves it but B fails to solve it?
7. Write difference between parameter and statistic.
8. What is sampling distribution? Construct frequency distribution table of sample mean in population 2, 4, 6, 8.
(i) Write down all possible sample size of two without replacement. (ii) Show that sample mean is an unbiased estimate of population mean.

Group C

Attempt any TWO questions [2x10=20]

9. What are different methods of measuring dispersion? Following are the marks of Basic Statistics obtained by two students A and B in 9 tests of 100 marks each.
Test: 123456789
Mark of A: 508273452580725662
Mark of B: 6073452580725662

B456555506862725256(i) Who is better? (ii) If the consistency of performance is the criteria for awarding a prize, who should get the prize?

- 10.** The mean weight of products is 68.22 grams with variance of 10.8 grams. How many products in a batch of 1000 would you expect (a) to be over 72 grams, (b) between 70 and 72 grams, and (c) below 65 grams?
- 11.** A part of the investigation of the collapse of the roof of a building, a testing laboratory is given all the available bolts that connected the steel structure at 3 different positions on the roof. The forces required to see each of these bolts are as follows: Position 1908279988391 Position 2958375102959092 Position 383898094 Perform an analysis of variance to test at the 0.05 level of significance whether the differences among the sample means at the three positions are significant.